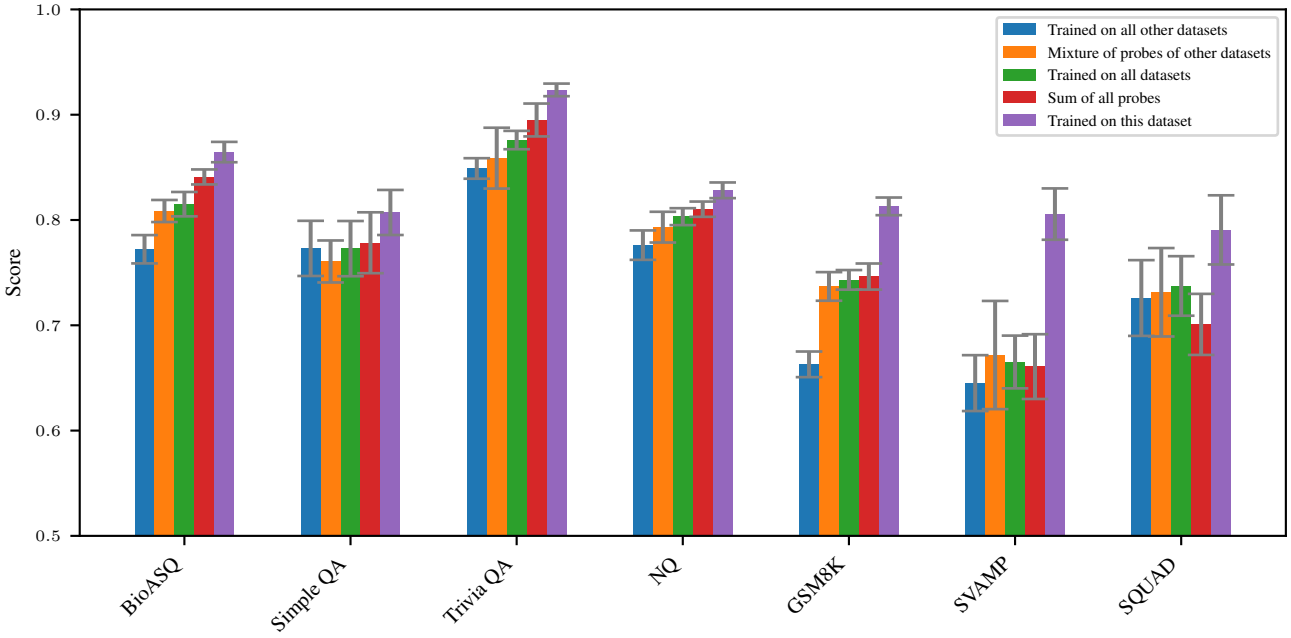

Figure 82. AUROC of linear probes at the stop token of the output in the multi-task setting, using L2 regularization.

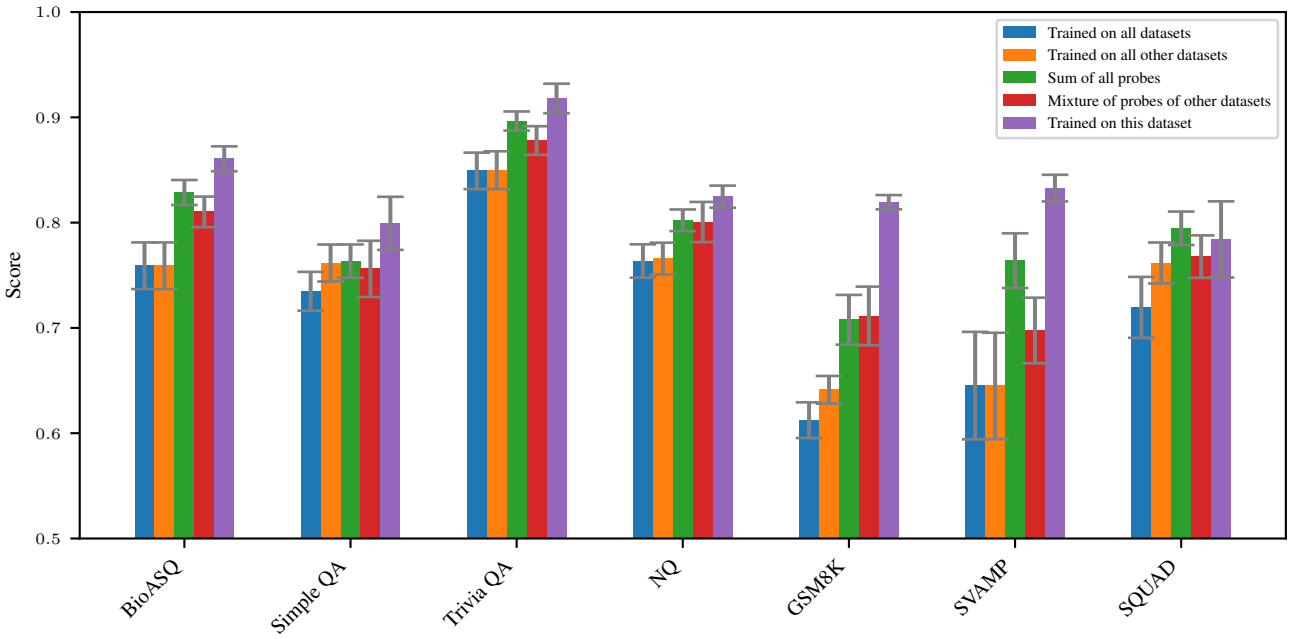

Figure 83. AUROC of linear probes at the token before the stop token of the output in the multi-task setting, using L2 regularization.

C. Phi-4 Mini Instruct

C.1. Layer 32

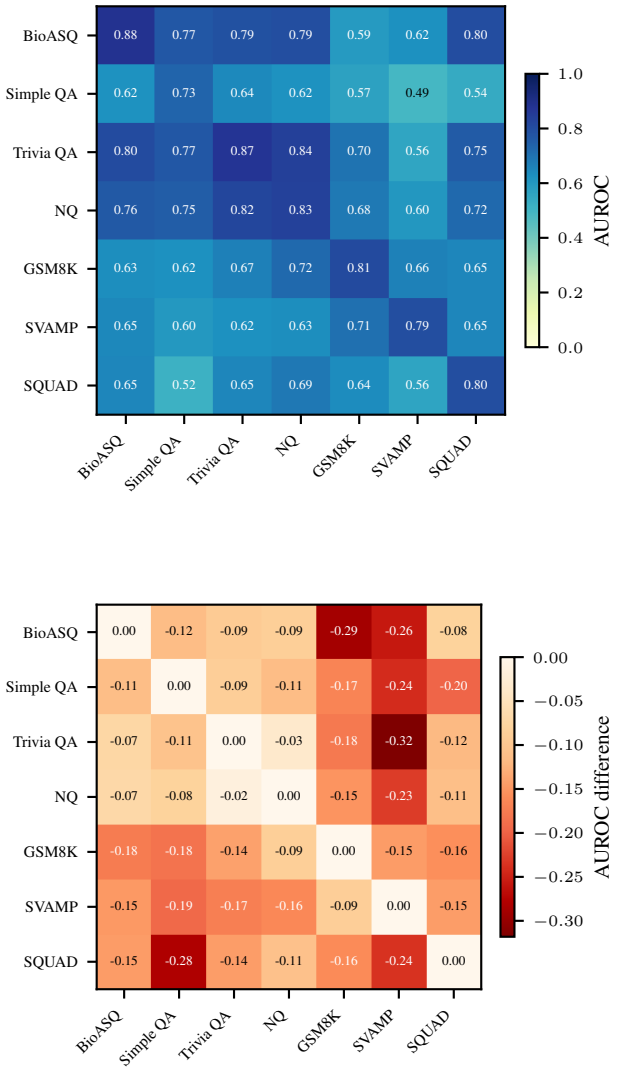

Figure 84. AUROC of probes trained on different tasks on the stop token of the output on last layer. Rows correspond to evaluation tasks while columns correspond to training tasks. The second plot represents the difference between the probe trained on this task and probes trained on the other datasets. Results are averaged over 5 runs.

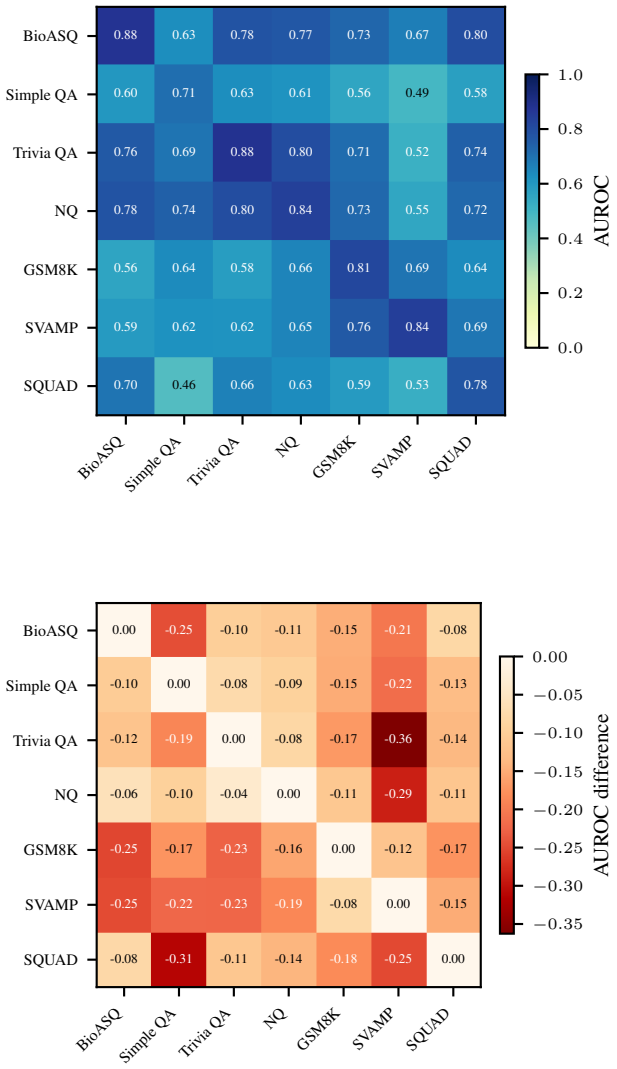

Figure 85. AUROC of probes trained on different tasks on the token before the stop token of the output on last layer. Rows correspond to evaluation tasks while columns correspond to training tasks. The second plot represents the difference between the probe trained on this task and probes trained on the other datasets. Results are averaged over 5 runs.

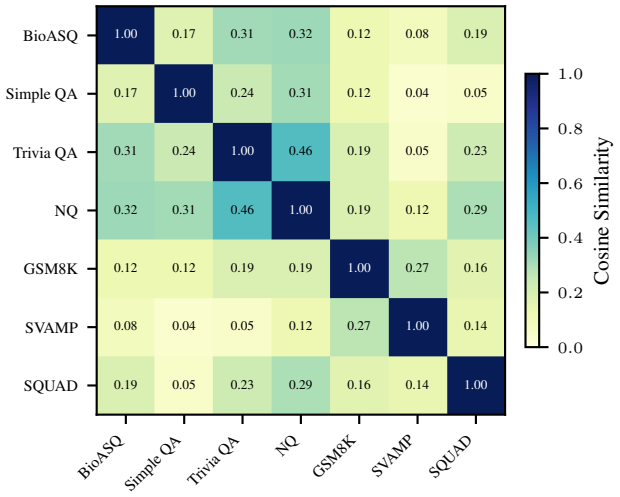

Figure 86. Cosine similarity between probes trained on different datasets using L2 regularization. Results are averaged over 5 runs.

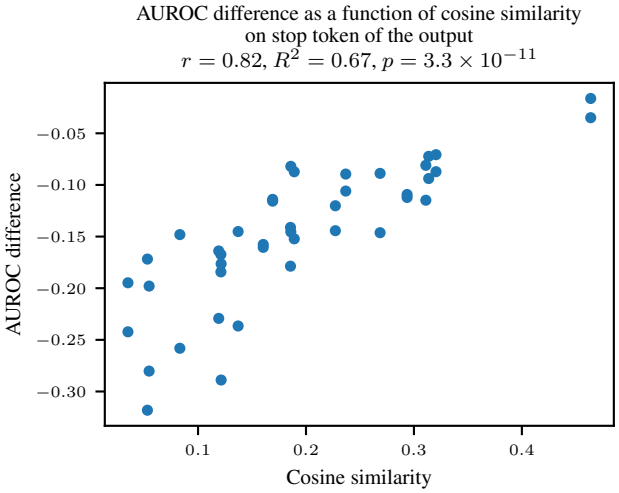

Figure 88. AUROC difference to probe trained on the right dataset as a function of cosine similarity between probes. Results are averaged over 5 runs.

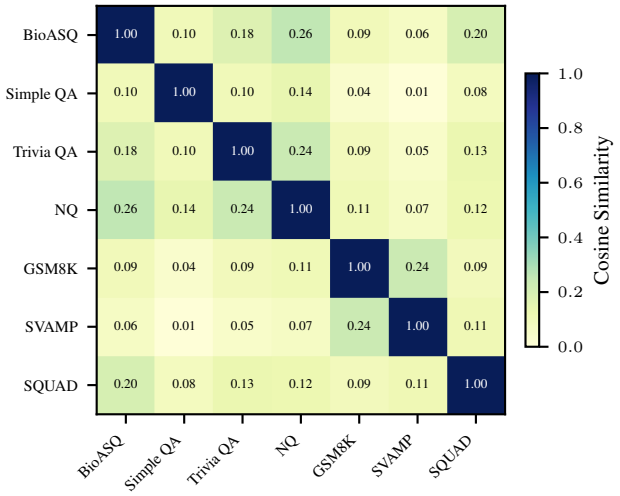

Figure 87. Cosine similarity between probes trained on different datasets using L2 regularization. Results are averaged over 5 runs.

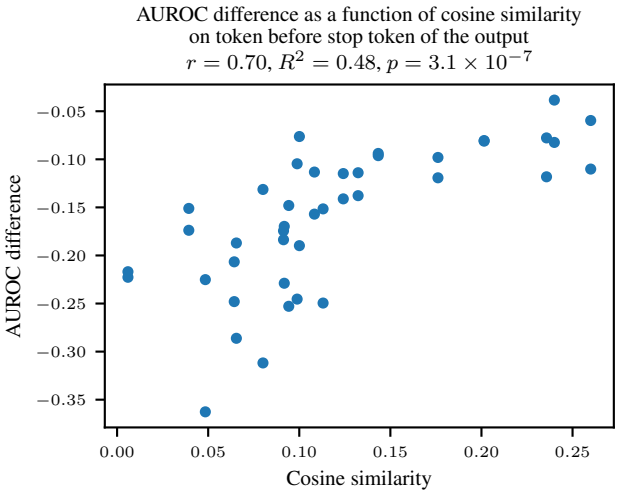

Figure 89. AUROC difference to probe trained on the right dataset as a function of cosine similarity between probes. Results are averaged over 5 runs.